

# STATISTICAL BUSINESS ANALYSIS REPORT

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## 1. Project Overview

### Introduction

Statistical analysis plays a vital role in understanding business performance and supporting data-driven decision-making. This project analyzes a sales dataset using various statistical techniques to identify trends, relationships, and patterns within the business data. The analysis includes descriptive statistics, data distribution analysis, hypothesis testing, confidence interval estimation, correlation analysis, and regression analysis.

The project was implemented using Python in Google Colab with several scientific computing libraries. Visualizations and statistical tests were used to interpret business performance and provide practical recommendations.

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### Project Goals

The primary goals of this project are:

- Analyze sales data using statistical methods.
- Calculate descriptive statistics for business variables.
- Study the distribution of sales data.
- Identify outliers using statistical techniques.
- Test the normality of numerical variables.
- Analyze relationships among sales variables.
- Perform hypothesis testing.
- Estimate confidence intervals.
- Build a linear regression model.
- Generate business insights and recommendations.

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## Project Objectives

The objectives of this project include:

- Calculate mean, median, mode, variance, and standard deviation.
  - Visualize data using histograms, box plots, density plots, scatter plots, and heatmaps.
  - Perform Shapiro-Wilk Normality Test.
  - Compute Pearson Correlation and Covariance.
  - Conduct three hypothesis tests:
    - One Sample T-Test
    - Independent T-Test
    - One-Way ANOVA
  - Calculate a 95% confidence interval.
  - Perform Linear Regression.
  - Evaluate model performance using the  $R^2$  Score.
  - Provide business recommendations based on statistical findings.
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## 2. Methodology and Results Interpretation

The statistical analysis was carried out in seven stages.

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### Day 1 – Descriptive Statistics

#### Methodology

Descriptive statistics summarize the main characteristics of the dataset.

The following measures were calculated for the numerical variables:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Minimum
- Maximum

- Range
- Quartiles
- Skewness
- Kurtosis

## Results Interpretation

- The **mean** provides the average value of each numerical variable.
- The **median** represents the central value and is less affected by extreme values.
- The **mode** identifies the most frequently occurring value.
- **Variance** and **standard deviation** indicate how widely the data is spread.
- **Skewness** shows whether the distribution is positively or negatively skewed.
- **Kurtosis** indicates whether the data distribution has heavier or lighter tails than a normal distribution.

These statistics provide an overall understanding of sales performance and variability.

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## Day 2 – Data Distribution Analysis

### Methodology

The distribution of numerical variables was analyzed using:

- Histograms
- Box Plots
- Density Plots
- Normal Probability Distribution
- Shapiro-Wilk Normality Test

Outliers were detected using the Interquartile Range (IQR) method.

### Results Interpretation

Histograms display the frequency distribution of each variable.

Box plots identify unusually high or low observations that may influence statistical analysis.

Density plots provide a smooth representation of the distribution.

The Shapiro-Wilk Test evaluates whether the data follows a normal distribution.

Decision Rule:

- $p\text{-value} > 0.05 \rightarrow$  Data is normally distributed.
- $p\text{-value} \leq 0.05 \rightarrow$  Data is not normally distributed.

The normality test determines whether parametric statistical methods are appropriate.

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## Day 3 – Correlation Analysis

### Methodology

Correlation analysis was performed using:

- Pearson Correlation Matrix
- Covariance Matrix
- Correlation Heatmap
- Pair Plot
- Scatter Plot

### Results Interpretation

The Pearson correlation coefficient measures the strength and direction of the relationship between two numerical variables.

Correlation values range from:

- $+1 \rightarrow$  Strong Positive Relationship
- $0 \rightarrow$  No Relationship
- $-1 \rightarrow$  Strong Negative Relationship

The heatmap visually represents correlation values using colors.

Scatter plots illustrate the relationship between Price and Total Sales.

Positive correlation indicates that both variables increase together, while negative correlation indicates an inverse relationship.

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## Day 4 – Hypothesis Testing

Three hypothesis tests were conducted.

### Hypothesis Test 1 – One Sample T-Test

### **Objective**

Determine whether the average Total Sales differs significantly from ₹150000.

### **Null Hypothesis ( $H_0$ )**

Mean Total Sales = ₹150000

### **Alternative Hypothesis ( $H_1$ )**

Mean Total Sales  $\neq$  ₹150000

### **Results Interpretation**

If the p-value is less than 0.05, the null hypothesis is rejected, indicating that the average Total Sales is significantly different from ₹150000.

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## **Hypothesis Test 2 – Independent T-Test**

### **Objective**

Compare the average sales between the North and South regions.

### **Null Hypothesis ( $H_0$ )**

Average sales are equal.

### **Alternative Hypothesis ( $H_1$ )**

Average sales are different.

### **Results Interpretation**

The test determines whether regional differences in sales are statistically significant.

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## **Hypothesis Test 3 – One-Way ANOVA**

### **Objective**

Compare average sales among all regions.

### **Null Hypothesis ( $H_0$ )**

All regional averages are equal.

### **Alternative Hypothesis ( $H_1$ )**

At least one region has a different average sales value.

### **Results Interpretation**

ANOVA evaluates whether regional sales differ significantly across multiple groups.

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## **Day 5 – Confidence Interval**

### **Methodology**

A 95% confidence interval was calculated using the sample mean and standard error.

Margin of Error was also computed.

### **Results Interpretation**

The confidence interval estimates the range in which the true population mean is expected to lie with 95% confidence.

The margin of error represents the maximum expected difference between the sample mean and the population mean.

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## **Day 6 – Regression Analysis**

### **Methodology**

Simple Linear Regression was performed.

Independent Variable:

- Price

Dependent Variable:

- Total Sales

The following metrics were calculated:

- Regression Coefficient

- Intercept
- $R^2$  Score
- OLS Regression Summary

## Results Interpretation

The regression equation predicts Total Sales based on Price.

The regression coefficient indicates how Total Sales changes with Price.

The intercept represents the predicted sales when Price equals zero.

The  $R^2$  Score measures how well the regression model explains the variation in Total Sales.

OLS regression provides detailed statistical information, including coefficient significance, confidence intervals, and model accuracy.

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## Day 7 – Business Insights

### Key Findings

- The descriptive statistics summarize business performance.
- Distribution analysis identifies variability and outliers.
- Correlation analysis reveals relationships among variables.
- Hypothesis testing validates statistical assumptions.
- Confidence intervals estimate future average sales.
- Regression analysis predicts Total Sales based on Price.

### Business Recommendations

- Increase inventory for high-demand products.
  - Focus promotional activities in high-performing regions.
  - Optimize pricing strategies using regression analysis.
  - Improve sales in low-performing regions through targeted marketing.
  - Continue monitoring sales trends using statistical techniques.
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## 3. Setup Instructions

## Step 1

Install Python 3.

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## Step 2

Install the required libraries.

```
pip install pandas  
pip install numpy  
pip install matplotlib  
pip install seaborn  
pip install scipy  
pip install scikit-learn  
pip install statsmodels
```

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## Step 3

Open Google Colab.

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## Step 4

Upload the following file:

sales\_data.csv

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## Step 5

Run all notebook cells sequentially.

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## Step 6

Review generated visualizations and statistical outputs.



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## 4. Code Structure

Statistical\_Business\_Analysis/

|

|— statistical\_analysis.ipynb

|— business\_data.csv

|— statistical\_report.pdf

|— hypothesis\_tests\_results.txt

|— requirements.txt

|— README.md

|

└─ visualizations/

    |— histogram.png

    |— boxplot.png

    |— density\_plot.png

    |— heatmap.png

    |— scatter\_plot.png

    └─ regression\_plot.png

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## 5. Visual Documentation

Include screenshots of the following outputs in the report:

- Dataset Preview

... Dataset Loaded Successfully!

|   | Date       | Product    | Quantity | Price | Customer_ID | Region | Total_Sales |
|---|------------|------------|----------|-------|-------------|--------|-------------|
| 0 | 2024-01-01 | Phone      | 7        | 37300 | CUST001     | East   | 261100      |
| 1 | 2024-01-02 | Headphones | 4        | 15406 | CUST002     | North  | 61624       |
| 2 | 2024-01-03 | Phone      | 2        | 21746 | CUST003     | West   | 43492       |
| 3 | 2024-01-04 | Headphones | 1        | 30895 | CUST004     | East   | 30895       |
| 4 | 2024-01-05 | Laptop     | 8        | 39835 | CUST005     | North  | 318680      |

- Dataset Information

```
=====
DATASET INFORMATION
=====
```

```
Shape of Dataset
(100, 7)
```

```
Columns
Index(['Date', 'Product', 'Quantity', 'Price', 'Customer_ID', 'Region',
      'Total_Sales'],
      dtype='object')
```

```
Data Types
Date          object
Product       object
Quantity      int64
Price         int64
Customer_ID   object
Region        object
Total_Sales   int64
dtype: object
```

```
Missing Values
Date          0
Product       0
Quantity      0
Price         0
Customer_ID   0
Region        0
Total_Sales   0
dtype: int64
```

```
Duplicate Rows
0
```

- Descriptive Statistics Output

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DESCRIPTIVE STATISTICS

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|        | Date                | Product | Quantity   | Price        | Customer_ID \ |
|--------|---------------------|---------|------------|--------------|---------------|
| count  | 100                 | 100     | 100.000000 | 100.000000   | 100           |
| unique | NaN                 | 5       | NaN        | NaN          | 100           |
| top    | NaN                 | Tablet  | NaN        | NaN          | CUST001       |
| freq   | NaN                 | 26      | NaN        | NaN          | 1             |
| mean   | 2024-02-19 12:00:00 | NaN     | 4.780000   | 25808.510000 | NaN           |
| min    | 2024-01-01 00:00:00 | NaN     | 1.000000   | 1308.000000  | NaN           |
| 25%    | 2024-01-25 18:00:00 | NaN     | 2.750000   | 14965.250000 | NaN           |
| 50%    | 2024-02-19 12:00:00 | NaN     | 5.000000   | 24192.000000 | NaN           |
| 75%    | 2024-03-15 06:00:00 | NaN     | 7.000000   | 38682.250000 | NaN           |
| max    | 2024-04-09 00:00:00 | NaN     | 9.000000   | 49930.000000 | NaN           |
| std    | NaN                 | NaN     | 2.588163   | 13917.630242 | NaN           |

|        | Region | Total_Sales   |
|--------|--------|---------------|
| count  | 100    | 100.000000    |
| unique | 4      | NaN           |
| top    | North  | NaN           |
| freq   | 28     | NaN           |
| mean   | NaN    | 123650.480000 |
| min    | NaN    | 6540.000000   |
| 25%    | NaN    | 39517.500000  |
| 50%    | NaN    | 97955.500000  |
| 75%    | NaN    | 175792.500000 |
| max    | NaN    | 373932.000000 |
| std    | NaN    | 100161.085275 |

=====  
Column : Quantity  
=====

Mean : 4.78  
Median : 5.0  
Mode : 4  
Variance : 6.698585858585861  
Standard Deviation : 2.588162641447763  
Minimum : 1  
Maximum : 9  
Range : 8  
25th Percentile : 2.75  
50th Percentile : 5.0  
75th Percentile : 7.0  
Skewness : 0.07297715917209194  
Kurtosis : -1.1999524820266472

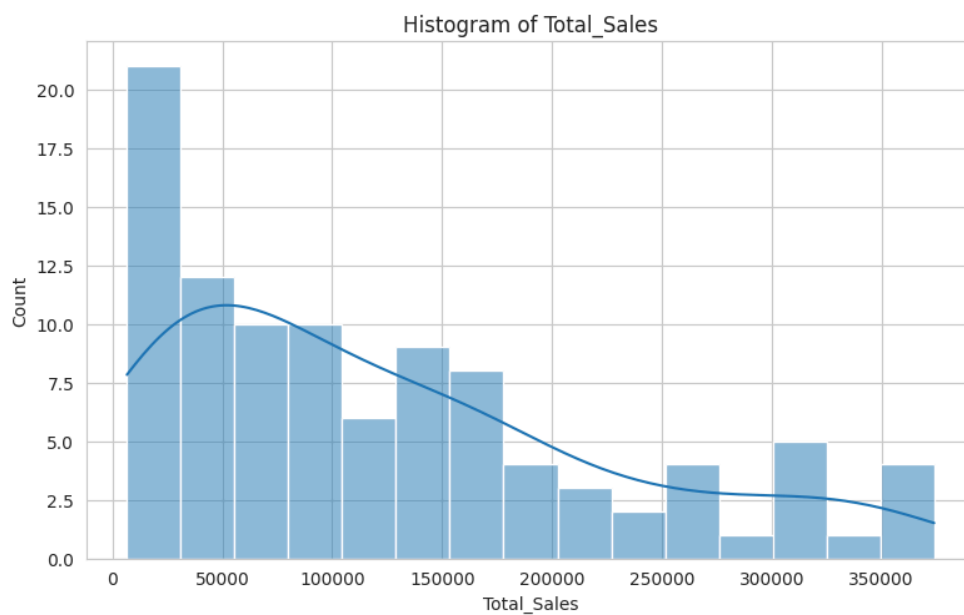
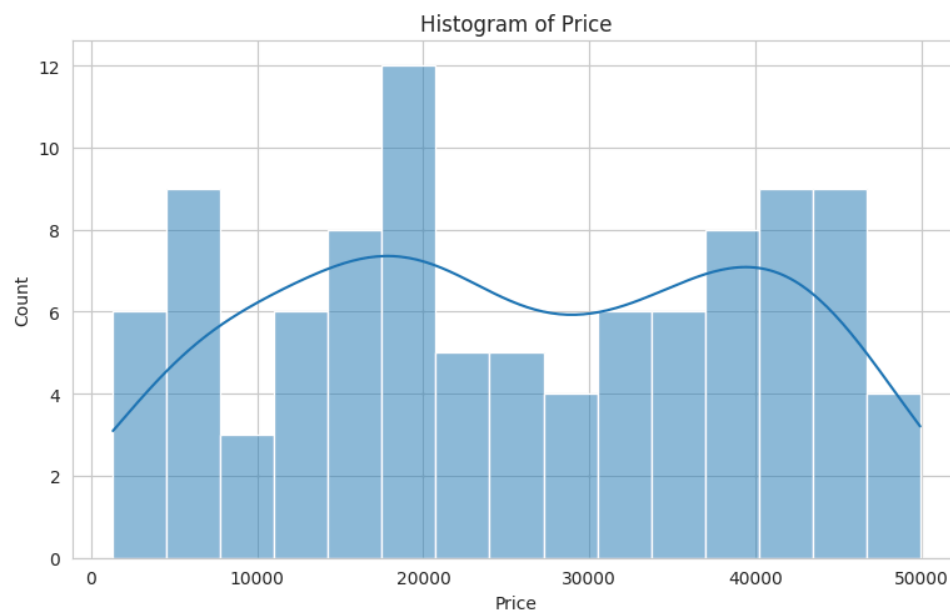
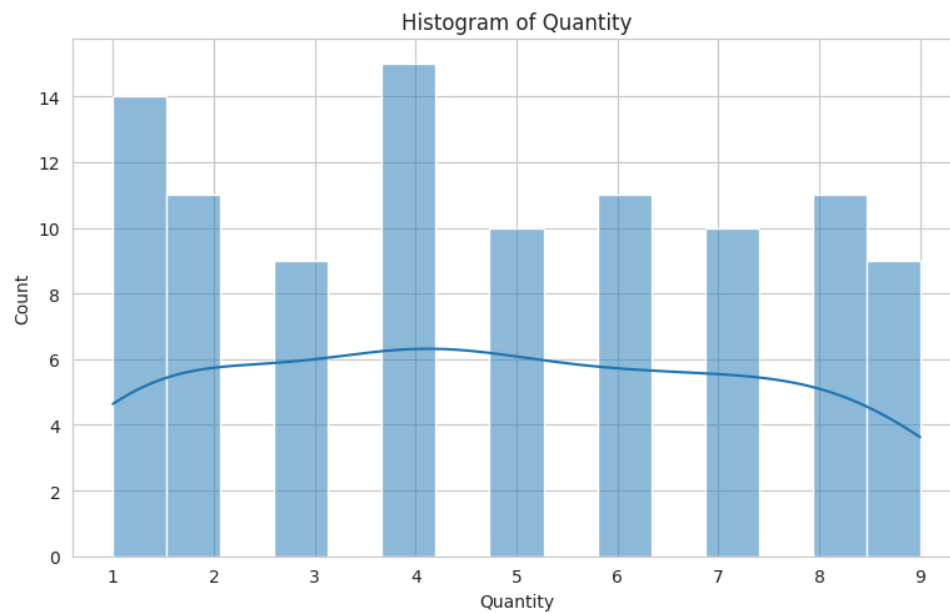
=====  
Column : Price  
=====

Mean : 25808.51  
Median : 24192.0  
Mode : 1308  
Variance : 193700431.54535356  
Standard Deviation : 13917.630241724111  
Minimum : 1308  
Maximum : 49930  
Range : 48622  
25th Percentile : 14965.25  
50th Percentile : 24192.0  
75th Percentile : 38682.25  
Skewness : -0.005833858155656331  
Kurtosis : -1.25911060832073

=====  
Column : Total\_Sales  
=====

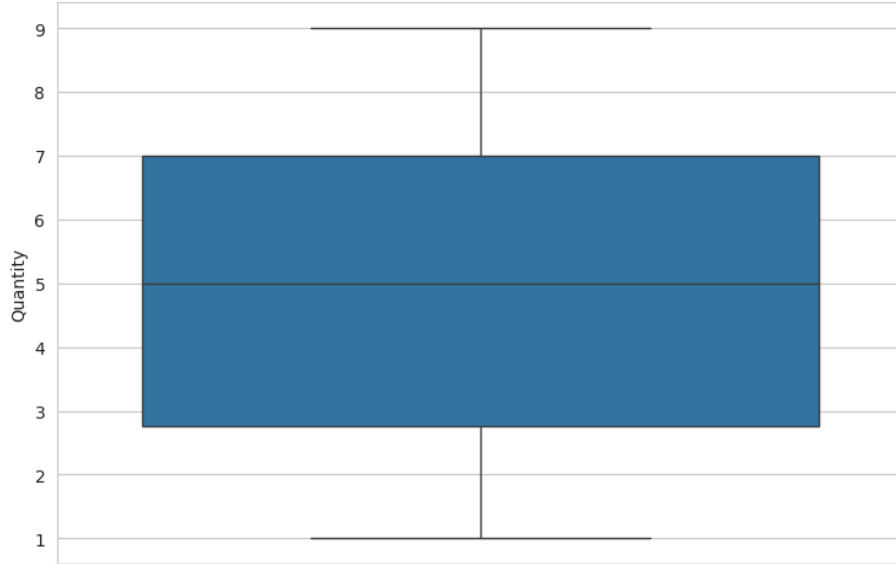
Mean : 123650.48  
Median : 97955.5  
Mode : 6540  
Variance : 10032243003.464245  
Standard Deviation : 100161.08527499213  
Minimum : 6540  
Maximum : 373932  
Range : 367392  
25th Percentile : 39517.5  
50th Percentile : 97955.5  
75th Percentile : 175792.5  
Skewness : 0.8784922285609769  
Kurtosis : -0.18281709321454587

- Histogram of Quantity, Price and Total Sales

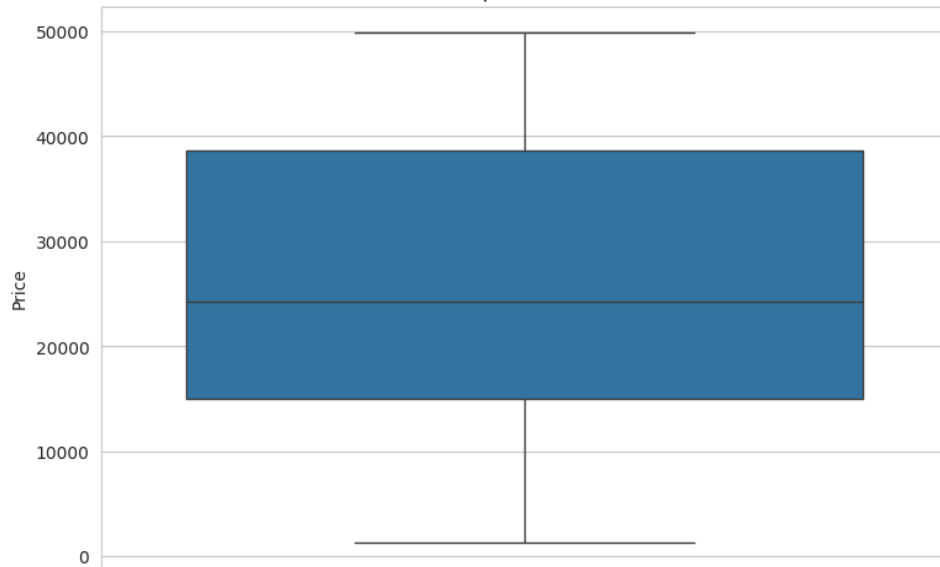


- Box Plots of Quantity, Price and Total Sales

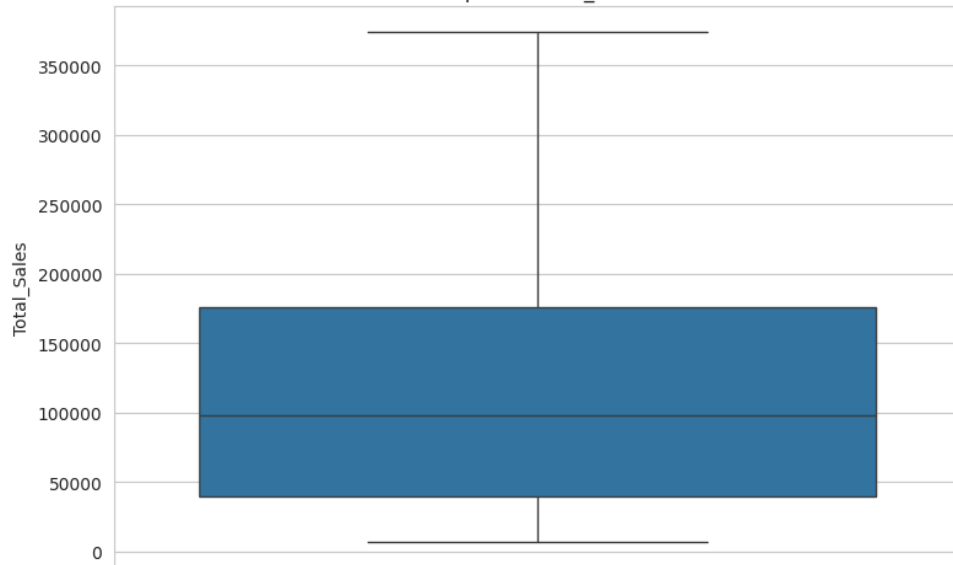
Boxplot of Quantity



Boxplot of Price

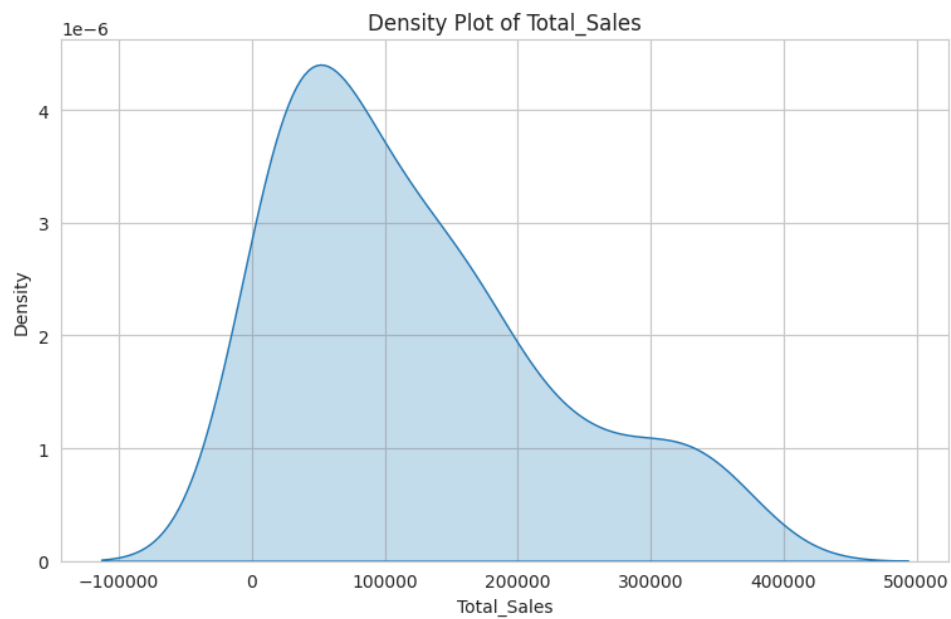
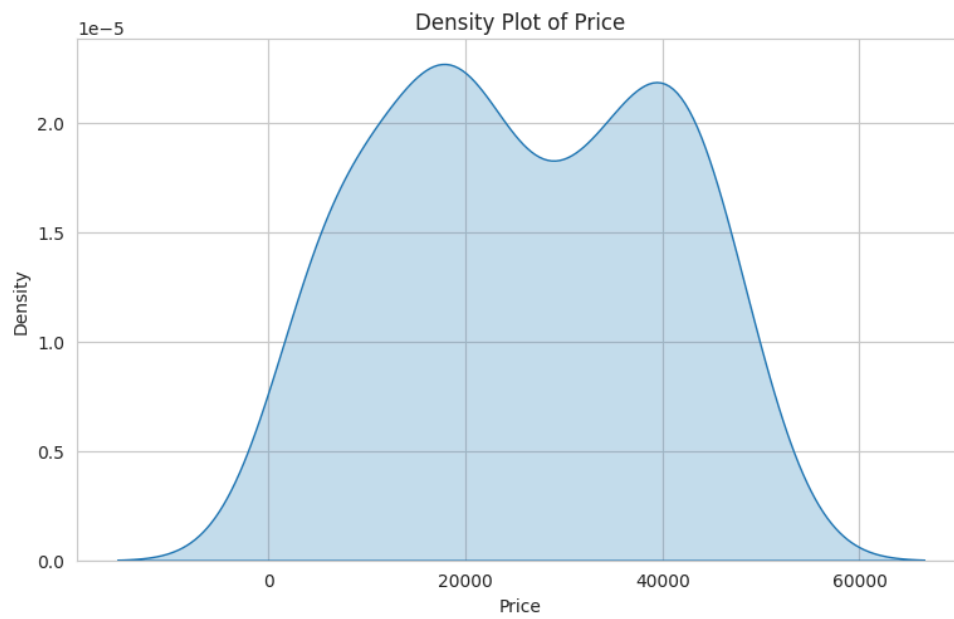
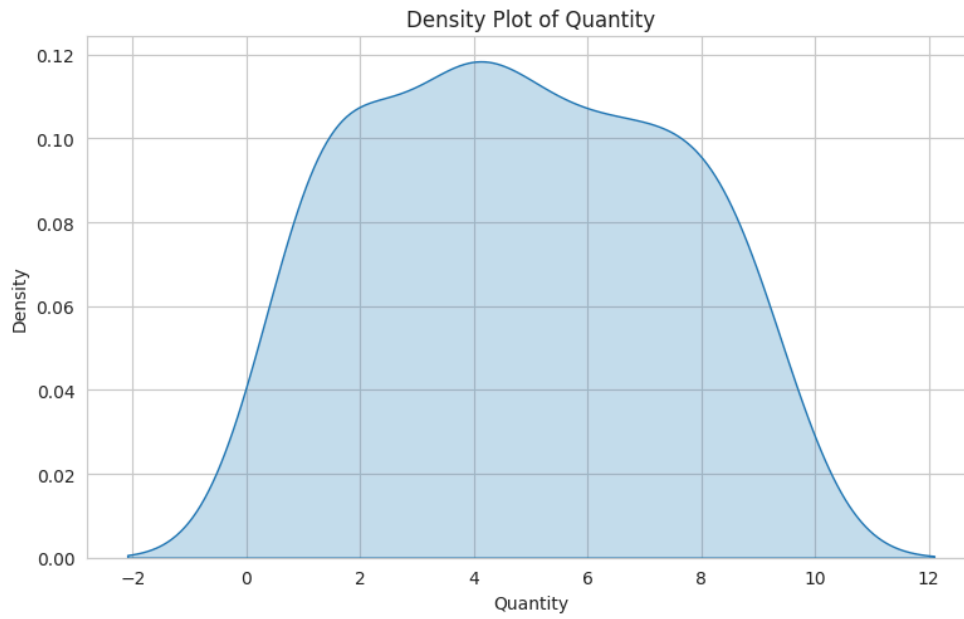


Boxplot of Total\_Sales

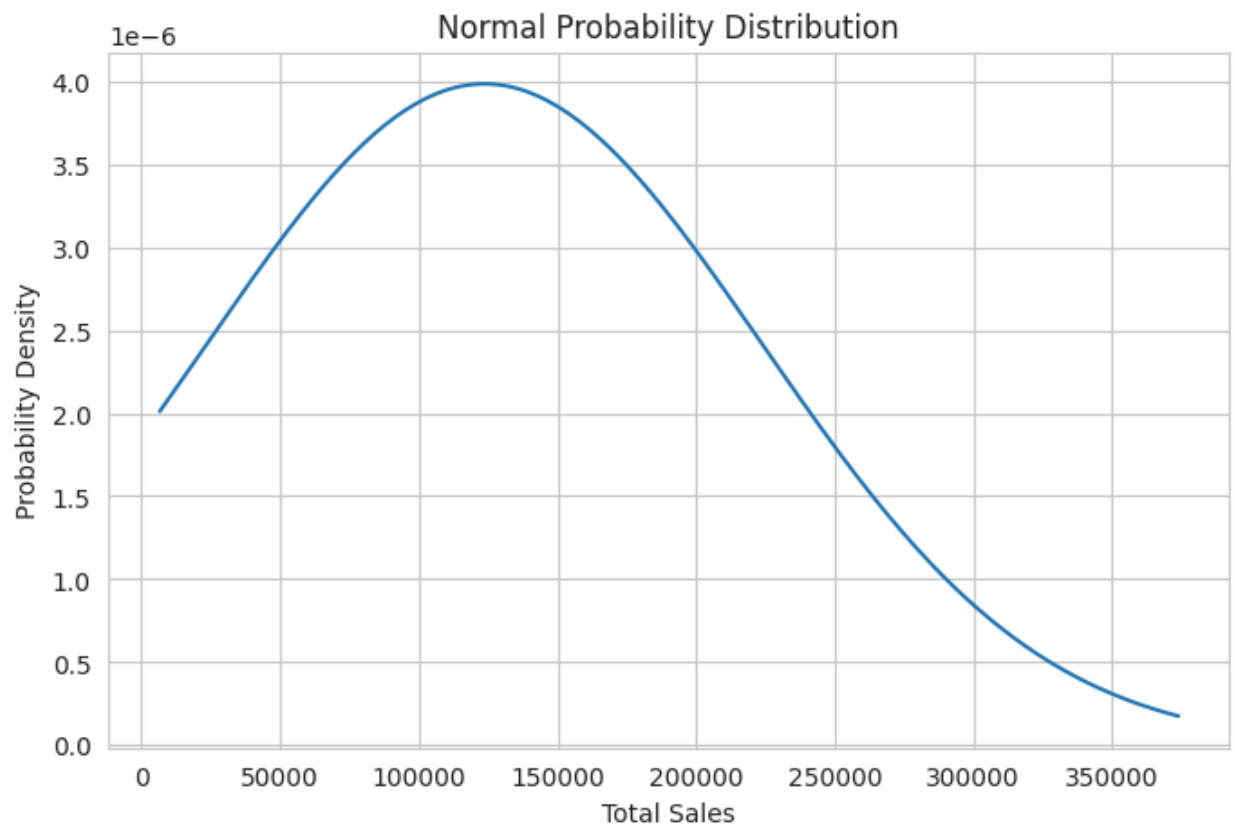


- Density Plots

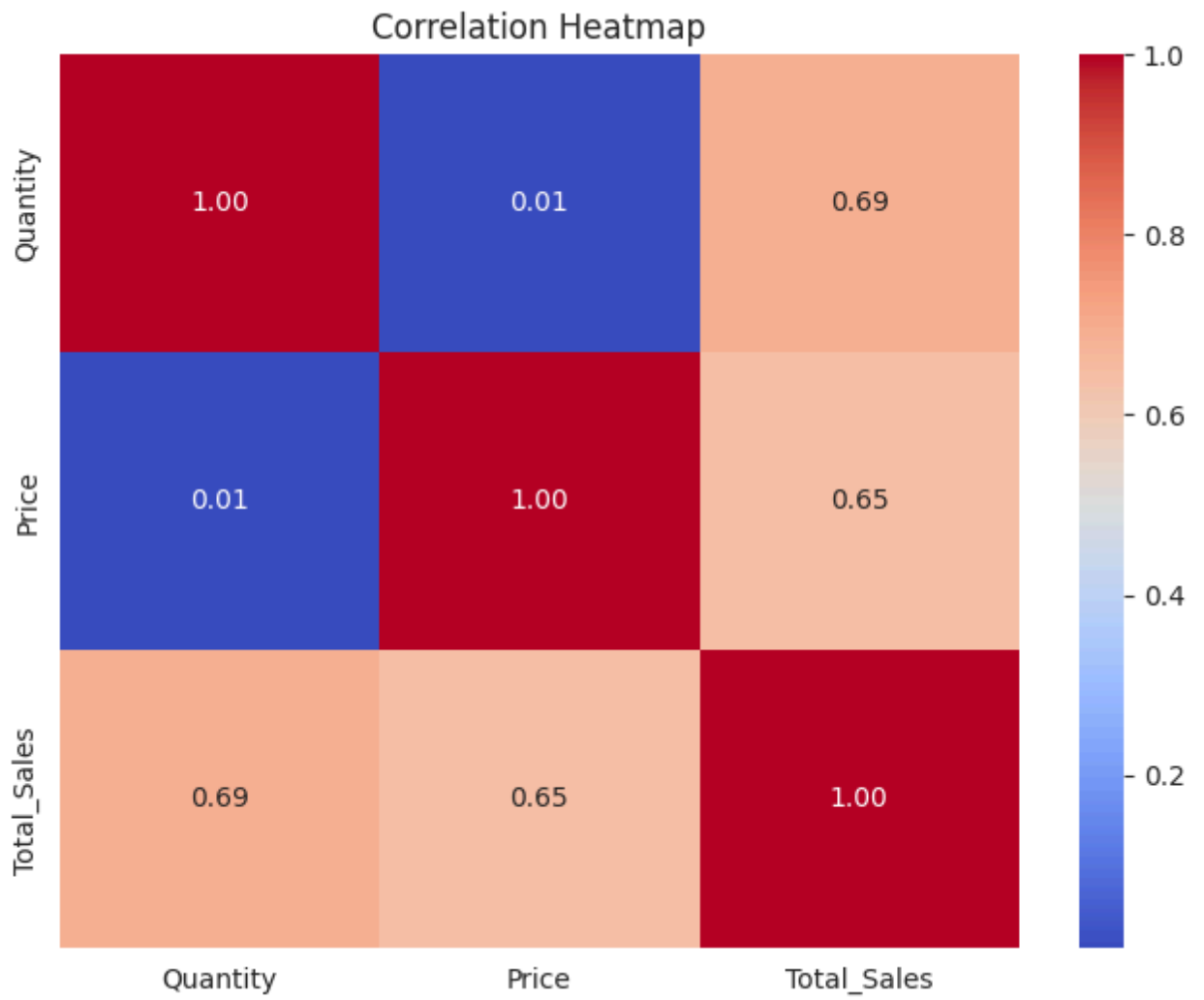




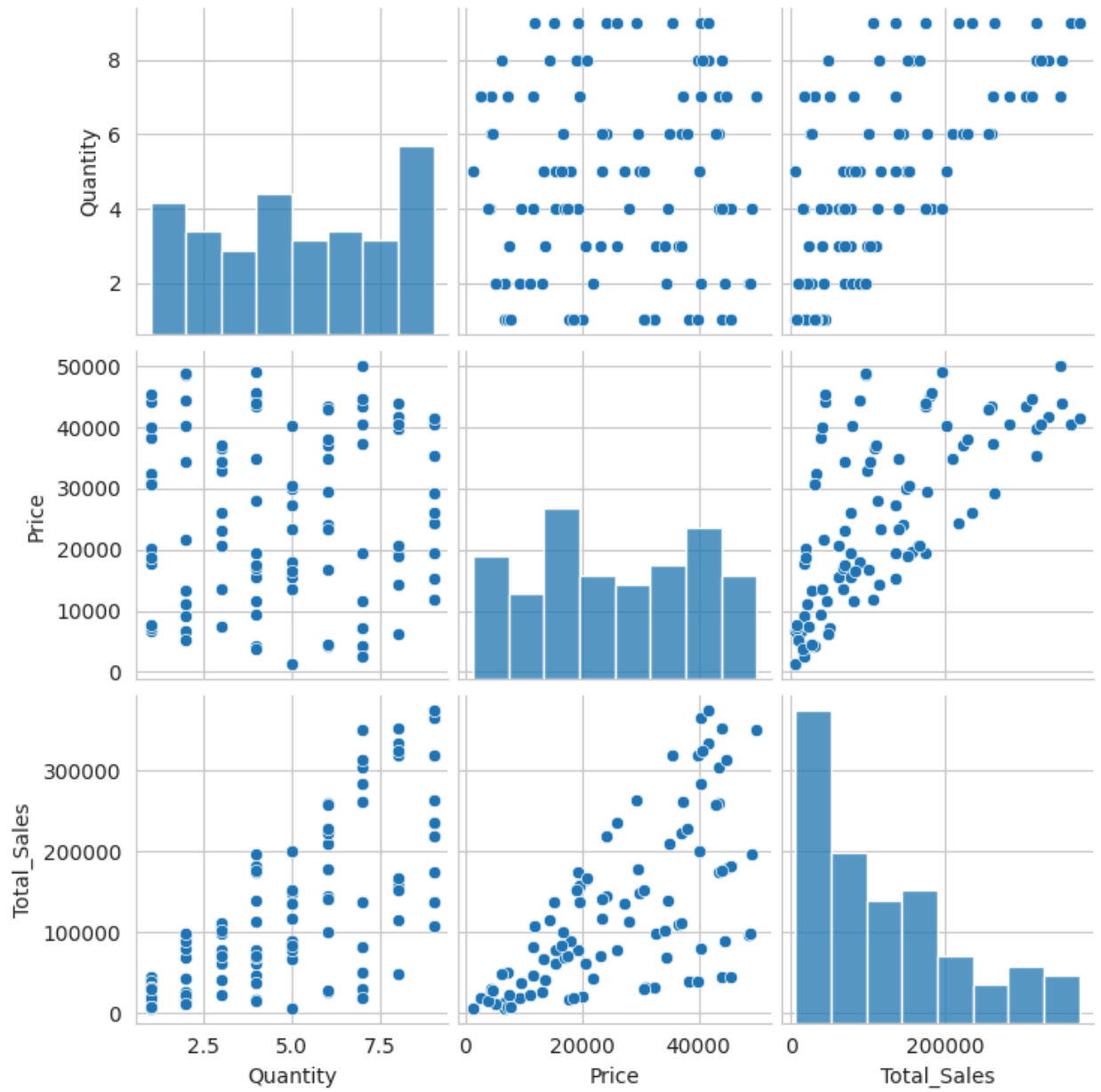
- Normal probability distribution



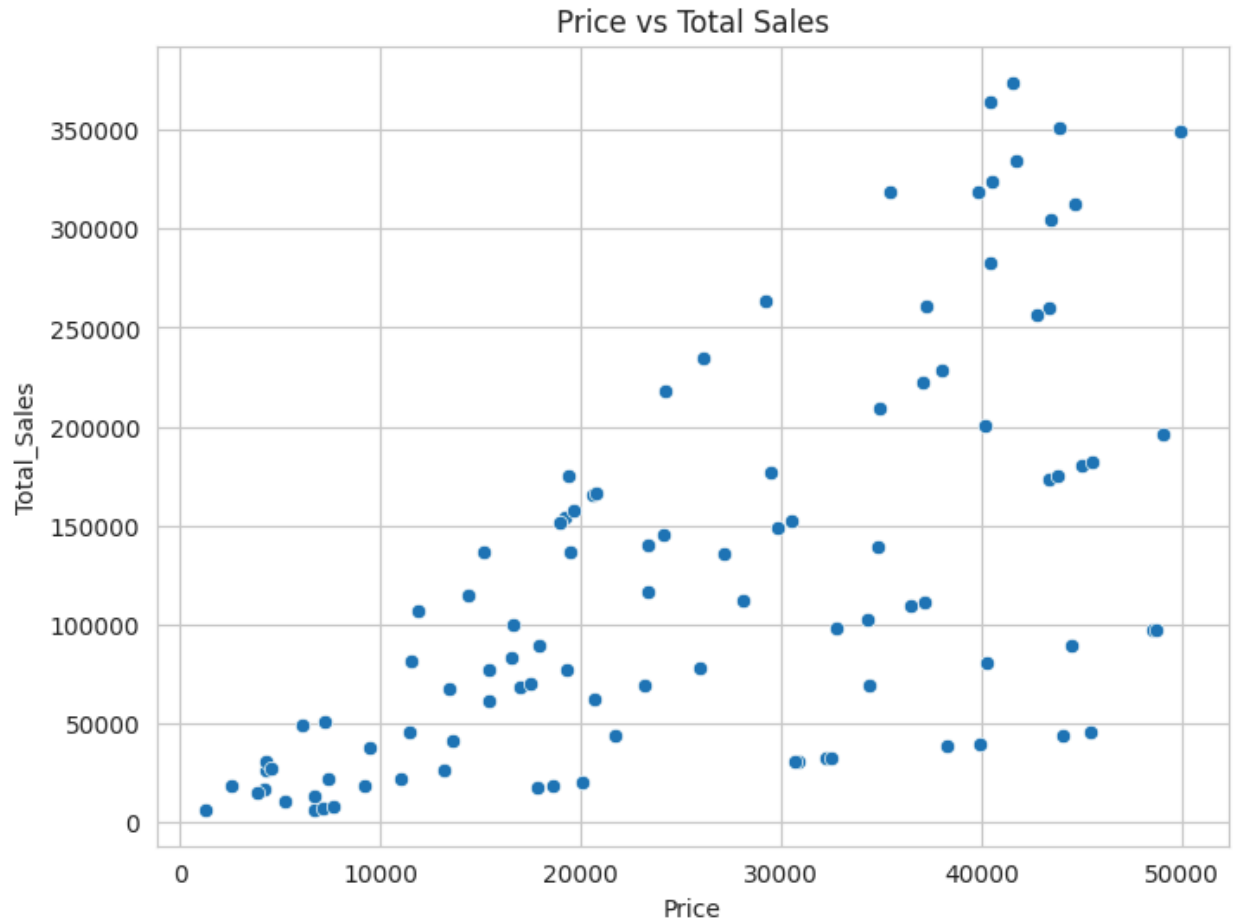
- Correlation Heatmap



- Pair Plot



- Scatter Plot



- OLS Regression Summary

```

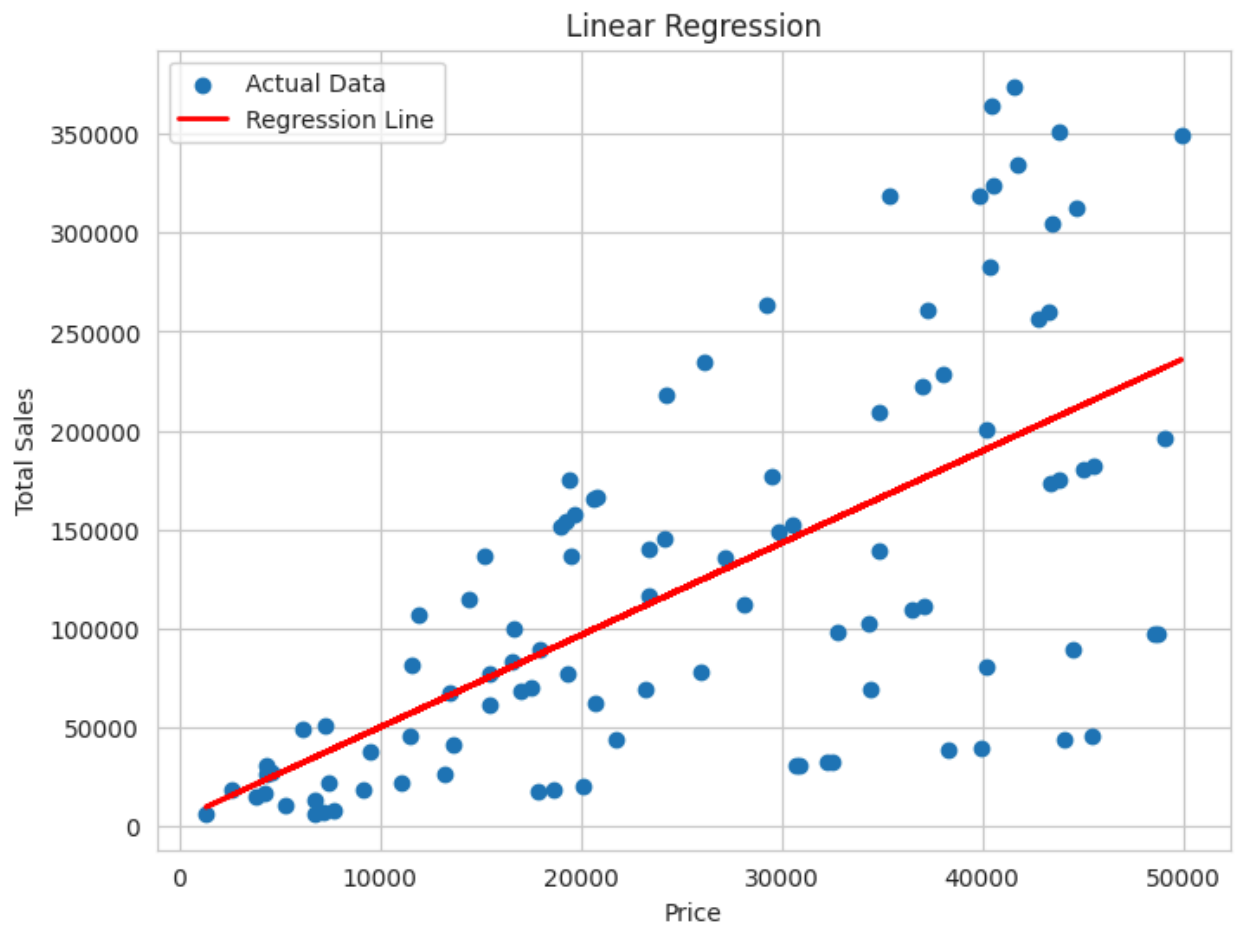
... =====
OLS REGRESSION SUMMARY
=====
                        OLS Regression Results
=====
Dep. Variable:          Total_Sales    R-squared:                0.417
Model:                  OLS           Adj. R-squared:           0.412
Method:                 Least Squares  F-statistic:              70.24
Date:                   Tue, 30 Jun 2026  Prob (F-statistic):      3.88e-13
Time:                   06:33:37       Log-Likelihood:          -1265.8
No. Observations:       100           AIC:                     2536.
Df Residuals:           98            BIC:                     2541.
Df Model:               1
Covariance Type:        nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const        3640.5435    1.63e+04     0.224     0.823    -2.86e+04    3.59e+04
Price         4.6500        0.555     8.381     0.000         3.549         5.751
=====
Omnibus:                0.059    Durbin-Watson:           2.169
Prob(Omnibus):           0.971    Jarque-Bera (JB):         0.183
Skew:                   0.048    Prob(JB):                 0.912
Kurtosis:               2.814    Cond. No.                  6.19e+04
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 6.19e+04. This might indicate that there are strong multicollinearity or other numerical problems.

- Regression Plot



- Final Business Insights Output

```
... =====
BUSINESS INSIGHTS
=====

1. Average Total Sales : 123650.48

2. Highest Performing Product
Product
Laptop      3889210
Tablet      2884340
Phone       2859394
Headphones  1384033
Monitor     1348071
Name: Total_Sales, dtype: int64

3. Highest Performing Region
Region
North      3983635
South      3737852
East       2519639
West       2123922
Name: Total_Sales, dtype: int64

4. Correlation Matrix
      Quantity    Price  Total_Sales
Quantity  1.000000  0.008014    0.688107
Price     0.008014  1.000000    0.646131
Total_Sales 0.688107  0.646131    1.000000
```

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## 6. Technical Details

### Programming Language

Python 3

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### Development Environment

Google Colab

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### Libraries Used

- Pandas
  - NumPy
  - Matplotlib
  - Seaborn
  - SciPy
  - Scikit-learn
  - Statsmodels
- 

## **Algorithms Used**

### **Descriptive Statistics**

Calculates central tendency and dispersion measures.

### **IQR Outlier Detection**

Identifies extreme observations using quartiles.

### **Shapiro-Wilk Test**

Tests whether numerical variables follow a normal distribution.

### **Pearson Correlation**

Measures the linear relationship between numerical variables.

### **Covariance Matrix**

Measures how two variables vary together.

### **One Sample T-Test**

Compares the sample mean against a known value.

### **Independent T-Test**

Compares the means of two independent groups.

### **One-Way ANOVA**

Compares the means of more than two groups.



## Confidence Interval

Estimates the population mean with a specified confidence level.

## Linear Regression

Models the relationship between Price and Total Sales.

## OLS Regression

Provides a detailed statistical summary of the regression model.

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## Data Structures

The project primarily uses **Pandas DataFrames** to store and manipulate the sales dataset. Numerical columns are processed using **NumPy arrays** during statistical computations, while dictionaries and lists are used internally for grouping data (for example, grouping sales by region for ANOVA).

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## Architecture

The project follows a sequential data analysis workflow:

1. Data Loading
2. Data Inspection and Cleaning
3. Descriptive Statistics
4. Data Visualization
5. Outlier Detection
6. Normality Testing
7. Correlation Analysis
8. Hypothesis Testing
9. Confidence Interval Estimation
10. Regression Analysis
11. Business Insights and Recommendations

This modular structure ensures each analysis stage builds logically on the previous one.

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## 7. Testing Evidence

| Test Case              | Expected Result                         | Actual Result                                                  | Status |
|------------------------|-----------------------------------------|----------------------------------------------------------------|--------|
| Dataset Loading        | Dataset loads successfully              | Dataset loaded successfully                                    | Pass   |
| Missing Value Check    | Missing values identified correctly     | Correct output displayed                                       | Pass   |
| Descriptive Statistics | Statistical measures calculated         | Mean, median, mode, variance, and standard deviation displayed | Pass   |
| Histogram Generation   | Histograms created                      | Histograms generated successfully                              | Pass   |
| Box Plot Generation    | Outliers visualized                     | Box plots displayed correctly                                  | Pass   |
| Shapiro-Wilk Test      | Test statistic and p-value returned     | Normality test completed                                       | Pass   |
| Correlation Analysis   | Correlation matrix generated            | Pearson correlation matrix displayed                           | Pass   |
| Heatmap                | Heatmap created                         | Heatmap generated successfully                                 | Pass   |
| Hypothesis Tests       | Test statistics and p-values calculated | One Sample T-Test, Independent T-Test, and ANOVA completed     | Pass   |
| Confidence Interval    | 95% confidence interval calculated      | Confidence interval and margin of error displayed              | Pass   |
| Regression Analysis    | Regression model fitted                 | Slope, intercept, $R^2$ score, and OLS summary generated       | Pass   |
| Business Insights      | Recommendations generated               | Business insights displayed successfully                       | Pass   |

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## 8. Conclusion

This project successfully demonstrates the application of statistical techniques to business sales data. Descriptive statistics provided a clear summary of the dataset, distribution analysis identified patterns and potential outliers, and correlation analysis highlighted relationships among key variables. Hypothesis testing validated statistical assumptions, confidence intervals estimated population parameters, and regression analysis modeled the relationship between price and total sales. The findings were translated into practical business recommendations, illustrating how statistical analysis can support informed business decisions and future sales planning.

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